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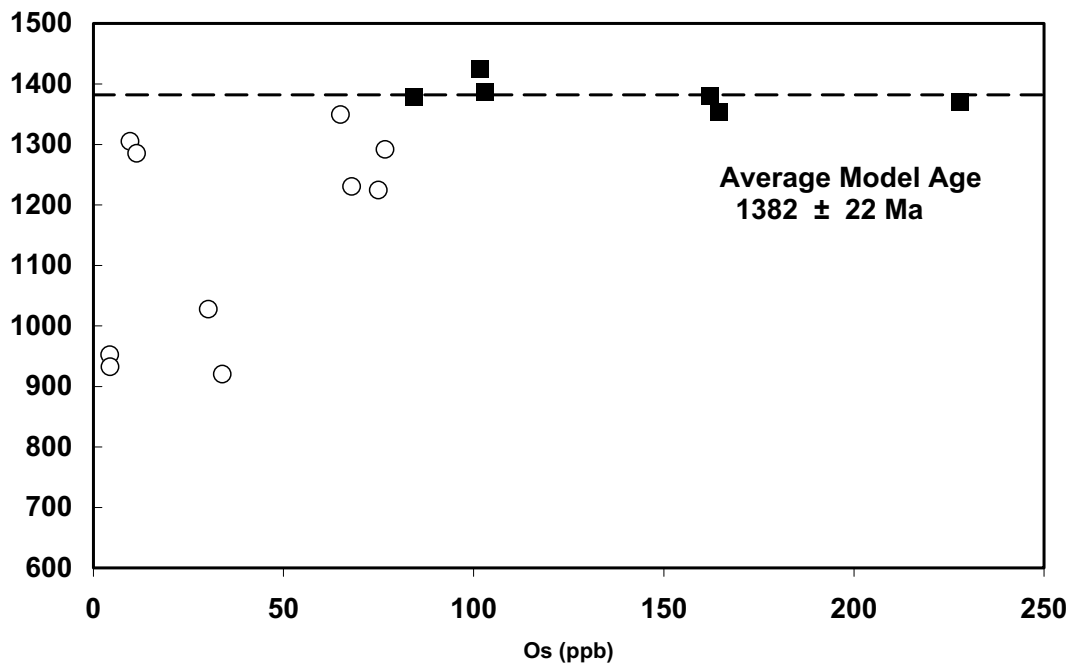


Figure 1. Variation of model ages of the Jinchuan Ni-Cu-PGE deposit with Os content of the samples.

Note that whereas samples with greater than 80 ppb Os (or Re/Os < 1.4) have uniform model ages, those with less than 80 ppb Os (or Re/Os > 1.4) have younger model ages. Some as bulk sulfide separates, although only 16 of the data sets for these samples are considered reliable. Samples were prepared for Re-Os isotope analysis in the ultra-clean Re-Os isotope laboratory at Monash University using well established isotope dilution techniques and sample dissolution in carius tubes. Final analyses were carried out on the VIEPS N-TIMS at La Trobe University, also in Melbourne. The Os contents of the bulk sulfide separates range from 162 to 228 ppb while their Re contents range from 198 to 282 ppb; this range in values in part reflects the varying purity of each of the separates. The range in Os contents (4.3 to 84.3 ppb) and Re contents (33.6 to 99.6 ppb) of the whole rock samples reflects the range in sulfide content of each of the samples. The very high Os contents of

the samples are characteristic of magmatic Ni-Cu-PGE sulfides and cumulate silicate rocks which formed from relatively high magnesium, primitive mantle-derived magmas.

When plotted on a $^{187}\text{Os}/^{188}\text{Os}$ vs $^{187}\text{Re}/^{188}\text{Os}$ isochron plot, the 16 reliable data define an isochron that yields an age of 877 ± 34 Ma and has an initial $^{187}\text{Os}/^{188}\text{Os} = 0.163 \pm 0.011$. Recalculation of the data excluding one sample produces an 15 point isochron that has an age of 879 ± 19 Ma, an initial $^{187}\text{Os}/^{188}\text{Os} = 0.165 \pm 0.006$, and a much lower MSWD (28 instead of 60). However, whereas samples with greater than 80 ppb Os (and Re/Os < 1.4) have model ages that are reasonably uniform at 1382 ± 22 Ma, those with Os contents less than 80 ppb (and Re/Os > 1.4) have model ages that are significantly less than this (Fig. 1). (The model age is the time at which the magma that formed the Jinchuan intrusive and its Ni-Cu-PGE ores was extracted from the mantle).

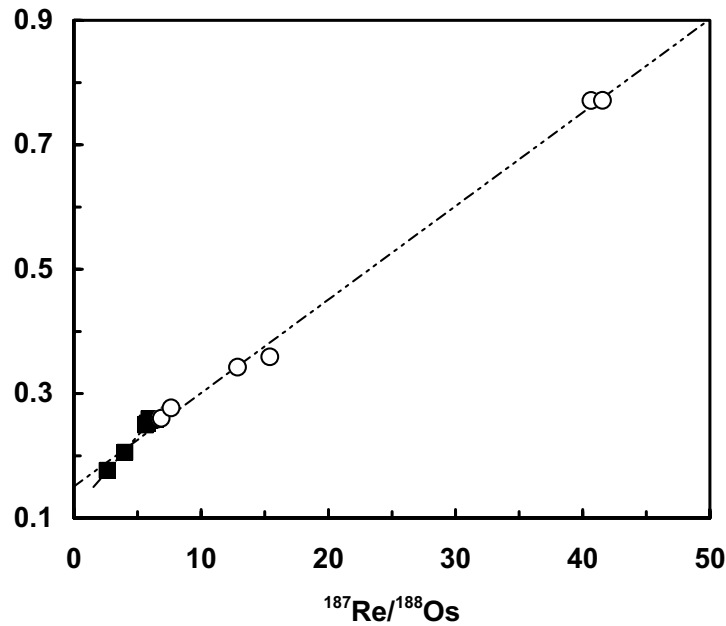


Figure 2. Isochron diagram for the Jinchuan samples. The slope of the isochron yields the age of samples that lie along it whereas the intercept of the isochron with the y axis yields the initial $^{187}\text{Os}/^{188}\text{Os}$ ratio of the magmas that formed the rocks. Samples with Re/Os ratios less than 1.4 define an isochron that yields an age of 1408 ± 140 Ma whereas samples with Re/Os ratios greater than 1.4 define an isochron that yields an age of 897 ± 66 Ma. See text for further details.

Samples with Re/Os ratios less than 1.4 (and Os contents > 80 ppb) define an isochron that yields an age of 1408 ± 140 Ma, an initial $^{187}\text{Os}/^{188}\text{Os}$ ratio of 0.113 ± 0.014 , and a MSWD of 20. On the other hand, samples with Re/Os ratios greater than 1.4 (and Os contents < 80 ppb) define an isochron that yields an age of 897 ± 66 Ma, an initial $^{187}\text{Os}/^{188}\text{Os}$ ratio of 0.150 ± 0.028 , and a MSWD of 128 (Fig. 2). The isochron age (1408 ± 140 Ma) for samples with Os > 80 ppb is within error of the model ages for these samples and is taken as the age of mineralization of the Jinchuan ultramafic body and its Ni-Cu-PGE ores. The large error on the isochron is due in part to the very narrow range in $^{187}\text{Re}/^{188}\text{Os}$ ratios of the samples that define the isochron. The initial $^{187}\text{Os}/^{188}\text{Os}$ ratio defined by the older,

primary isochron (0.113 ± 0.014) is very close to that of the mantle (0.12) and provides strong support that the magmas that formed Jinchuan were derived from the mantle and were NOT subjected to a significant amount of crustal contamination (a common triggering event for the formation of Ni-Cu-PGE sulfide ores). The initial $^{187}\text{Os}/^{188}\text{Os}$ ratio defined by the younger, secondary isochron (0.150 ± 0.028) is more radiogenic and carries a crustal signature (crustal rocks formed 1.8 Ga ago have an initial $^{187}\text{Os}/^{188}\text{Os}$ ratio = 1.8).

It is suggested the Jinchuan ultramafic body was subjected to an event at ~ 877 Ma that resulted in crustal fluids carrying some Re (derived from crustal rocks) but no Os permeating and interacting with the Jinchuan rocks. The Re-Os isotope systematics of

samples with a high Os content were relatively unaffected by the addition of the crustal Re. However, the Re-Os isotope systematics of samples with low Os contents were upset (or “re-set”) as relatively much more Re was added to these samples. The fact that two of the low Os content samples do not have much younger model ages suggests that fluid access was not uniform to all of Jinchuan samples.

In summary, the age of Ni-Cu-PGE sulfide mineralization of the Jinchuan ore system is 1408 Ma, but the Re-Os isotope system was re-set in low Os samples at ~ 877 Ma when metamorphic fluids carrying crustally-sourced Re permeated the Jinchuan ultramafic body. The disturbance at ~ 877 Ma may have been triggered by the breakup of the supercontinent Rodinia which may have been initiated by a mantle plume at 828 ± 7 Ma (Li et al., 1999). This event corresponds to the intrusion of mafic and ultramafic rocks as well as widespread granites in South China (Li et al., 1999).

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